Johnny Chau

Software Engineer · TypeScript / Node.js · Electronic Engineering

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TypeScript backend engineer working on real-time data systems, API integrations and AI-related applications. Recent projects include exchange integrations and processing of WebSocket fills and hedge orders. HKUST electronic engineering graduate with two years of postgraduate BioMEMS research and three co-authored publications.

TECHNICAL SKILLS

Software	TypeScript, JavaScript, Node.js, backend development, REST APIs, WebSockets, asynchronous processing, testing and browser automation.
AI coding & agents	Codex, Cursor, GitHub Copilot, local LLMs for coding and general-purpose agents.
Engineering	COMSOL Multiphysics, finite-element simulation, MEMS design and fabrication, microfluidics, embedded C, MATLAB, Simulink and control systems.
Languages	Cantonese, Mandarin, English.

PROFESSIONAL EXPERIENCE

PERIOD	EMPLOYER / ROLE	WORK
Dec 2023 – Present	**Hong Kong Kyowa Bussan Co., Ltd.** 香港共和物産株式会社 · AI Programmer, Japan	Develop and maintain TypeScript backend services for AI-assisted multilingual product descriptions and e-commerce market analysis.
Dec 2022 – Oct 2023	**劉昇通商株式会社** Software Engineer · TypeScript backend, Japan	Developed backend application features in TypeScript.
Jun 2021 – Nov 2022	**東京人材派遣株式会社** Software Engineer · TypeScript backend, Japan	Worked on server-side application development in TypeScript.
2018 – 2019	**HIGHTECH · HK Science Park** Application Engineer	MEMS Hall-sensor FEM simulation; C firmware development; coordination between engineering teams.
2015 – 2017	**HKUST** Postgraduate Researcher · Electronic Engineering	BioMEMS, microfluidic cell separation, COMSOL simulation, MEMS fabrication and clean-room work.
2015 – 2017	**HKUST** Teaching Assistant · Concurrent with research	Supported undergraduate electronics courses.

EDUCATION AND RESEARCH

Hong Kong University of Science and Technology (HKUST)

BEng in Electronic Engineering · First Class Honours · 2012–2015

Signal processing, control systems and robotics, including quadcopter work with MATLAB and Simulink.

Postgraduate Research — Electronic Engineering, HKUST

2015–2017

Microfluidics, dielectrophoretic cell sorting, BioMEMS, COMSOL finite-element simulation and microfabrication.

PUBLICATIONS

Railing cells along 3D microelectrode tracks for continuous-flow dielectrophoretic sorting

X. Xing, C. N. Ng, M. L. Chau, L. Yobas. *Lab on a Chip* 18(24), 3760–3769 (2018). DOI: 10.1039/C8LC00805A

Numerical simulation for cell railing on 3D silicon microelectrodes featuring sidewall undercuts

M. L. Chau, X. Xing, L. Yobas. MicroTAS 2017, pp. 822–823. Conference program

Dielectrophoretic cell sorting via sliding cells on 3D silicon microelectrodes

X. Xing, M. L. Chau, K. A. Roshan, L. Yobas. IEEE MEMS 2017, pp. 147–150. DOI: 10.1109/MEMSYS.2017.7863362